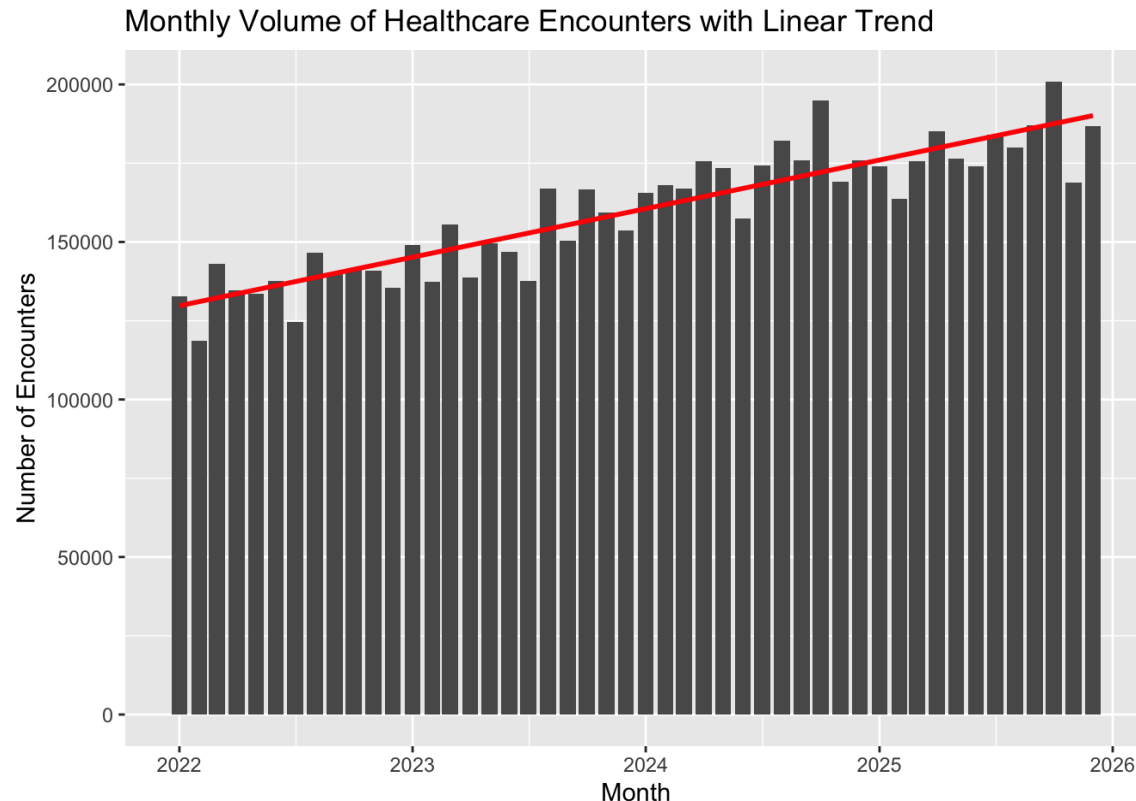


Post-Pandemic Growth and Unequal Acceleration

A Time Series Analysis of Patient Encounter Dynamics Across a Regional Health System In Kansas(2022–2025)

The Big Data Theory: Wenqi Cai, Qiaochu Chen, Peiran Li, Claire Xu, and Le Yu

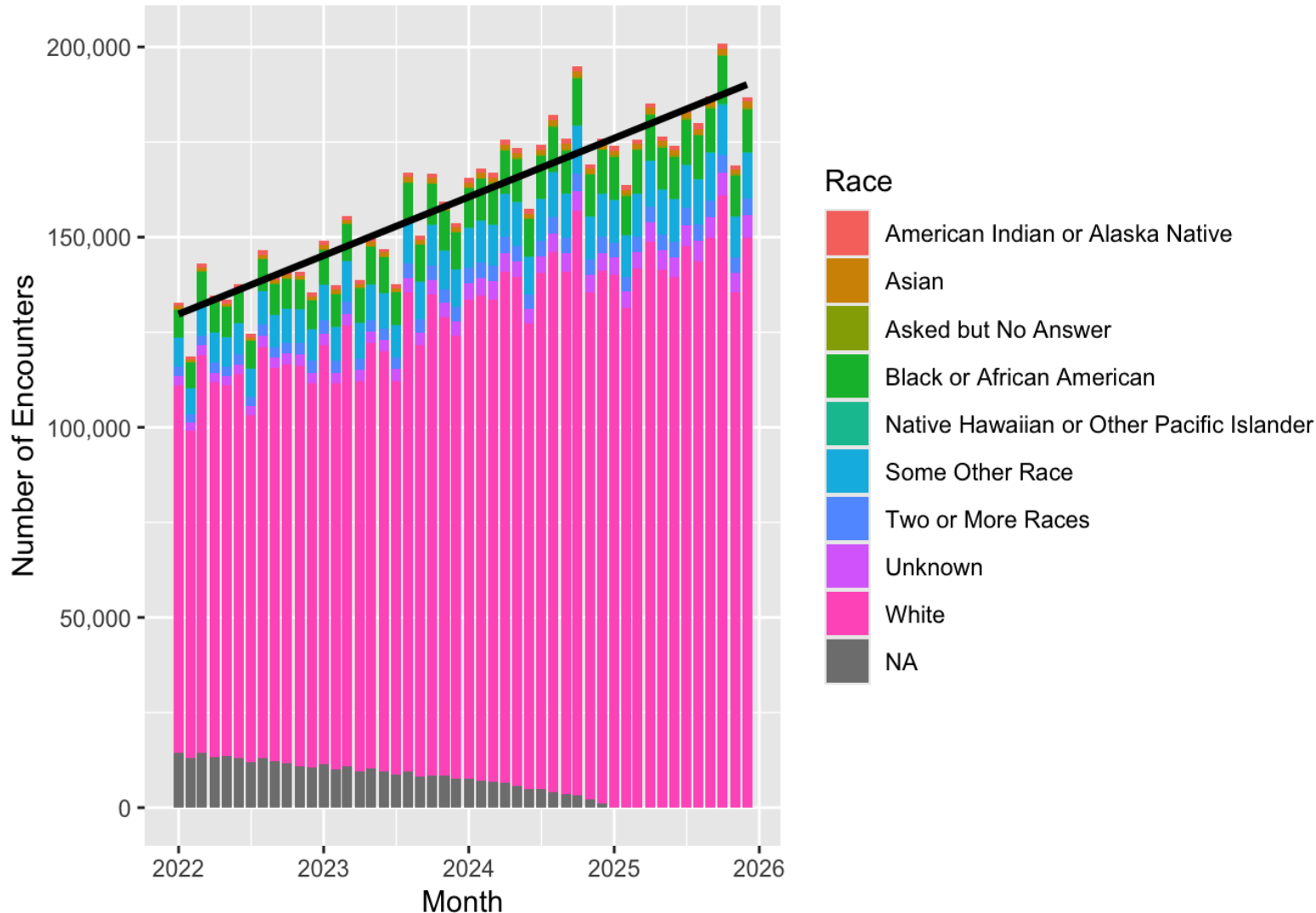
Motivation & Journey: Why this is important?



- **Monthly healthcare encounters have steadily increased over the past five years, reaching nearly 200,000 at peak.**
- **Rising volume makes it harder to identify patients in need and can obscure meaningful clinical patterns.**
- **However, encounter frequency also provides an opportunity to detect risk earlier and improve patient outcomes.**

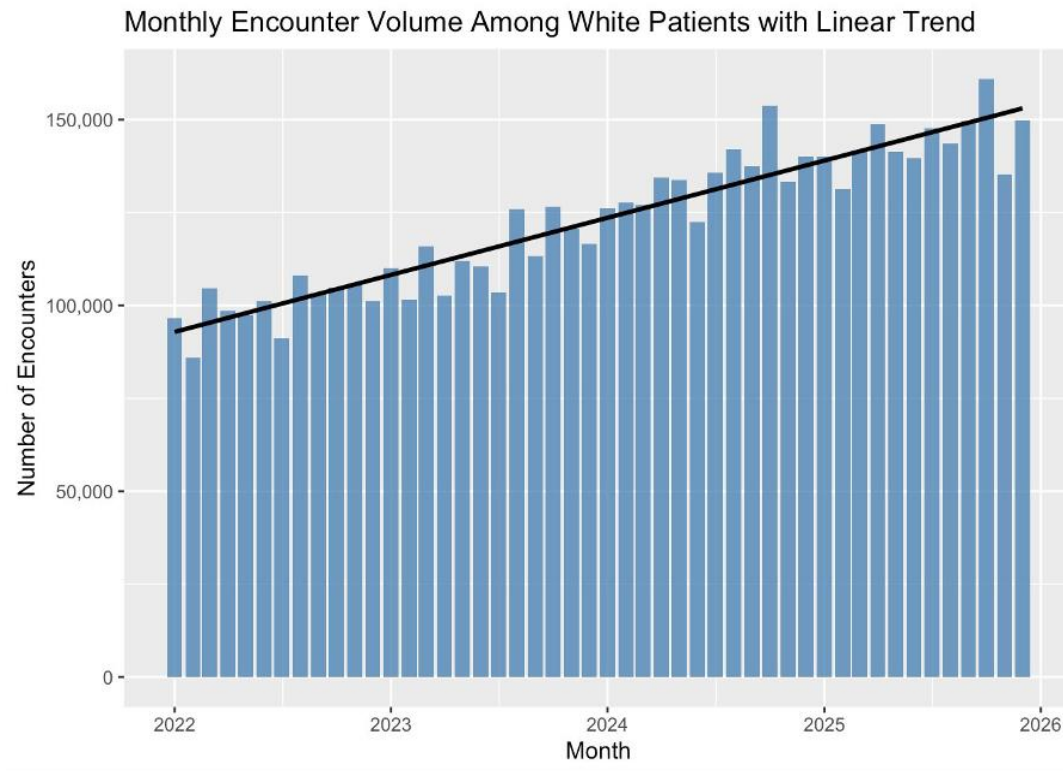
Encounter utilization is dominated by White patients, with consistent growth across all racial groups

Monthly Encounter Volume by Race with Linear Trend



- **White patients are overrepresented relative to the Topeka population baseline (~69%).**
- **Total encounter volume increases over time, indicating rising healthcare utilization.**
- **Missing race data (NA) declines to near zero, reflecting improved data quality.**
- **Utilization trends are parallel across racial groups, suggesting structural differences are driven by scale.**

White Patients Nearly Identical Growth to the Overall System

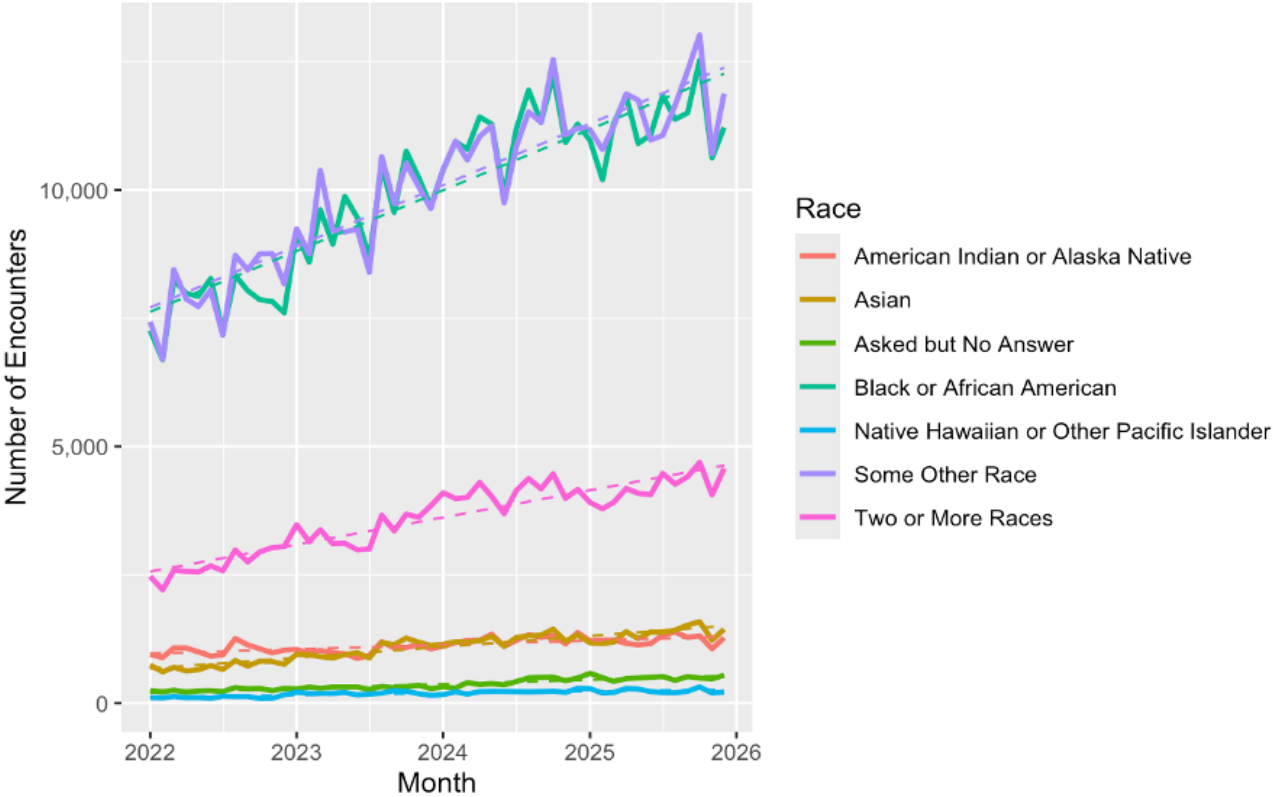


- White patients show an average monthly encounter increase of 1,281.11, nearly identical to the system average of 1,285.09.
- System-level growth is almost entirely mirrored by White patient utilization trends.
- This alignment contributes to White patients maintaining an increasing share of total encounters over time.



Non-White Growth: Positive, but Substantially Smaller in Magnitude

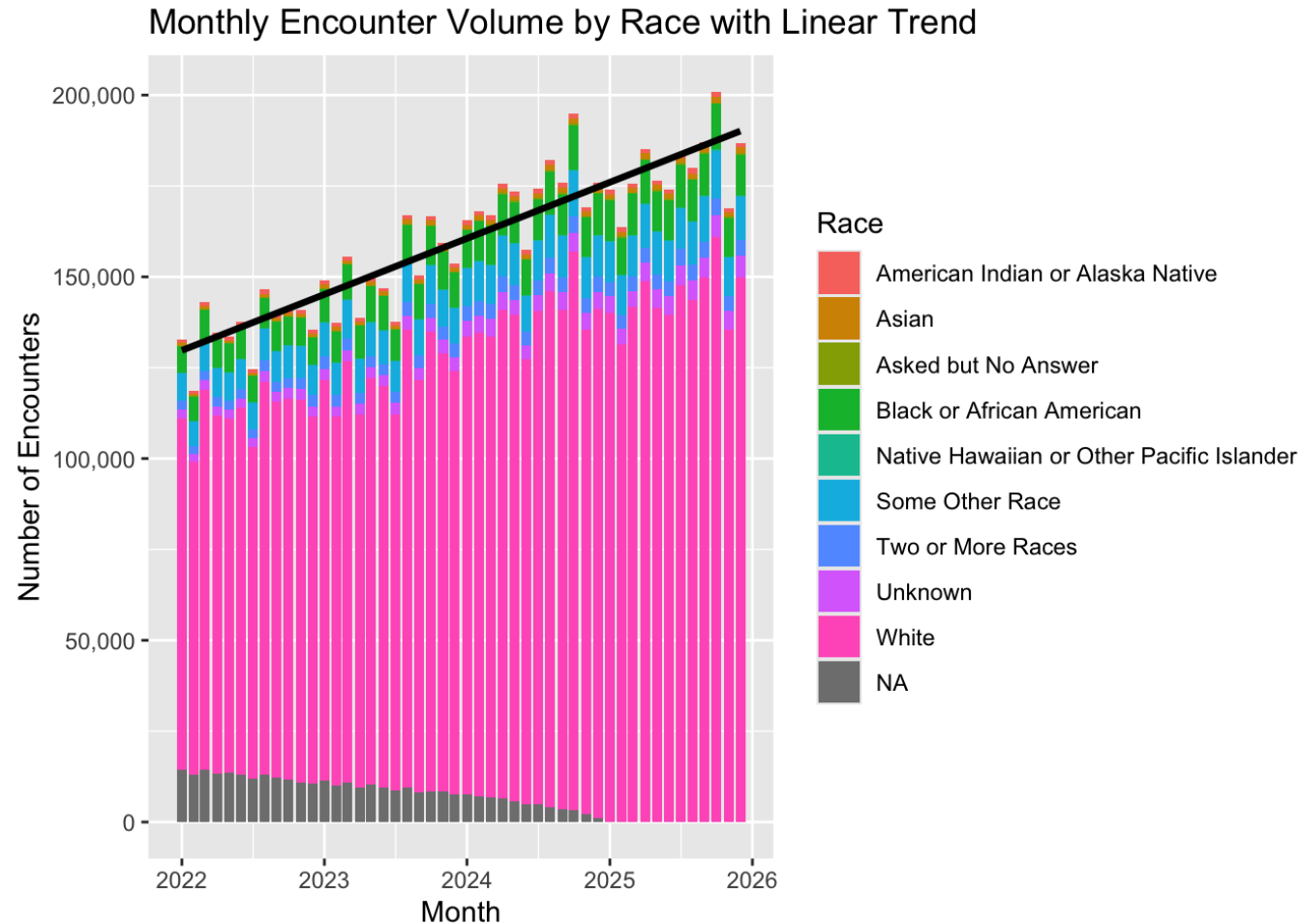
Monthly Encounter Volume Among Non-White Patients by Race
Dashed lines indicate linear trends



Race	Average Increasing Rate
American Indian or Alaska Native	7%
Asian	17.7%
Black or African American	98.8%
Native Hawaiian or Other Pacific Islander	3.1%
Some Other Race	99.5%
Two or More Races	44%
Asked but No Answer	6.9%

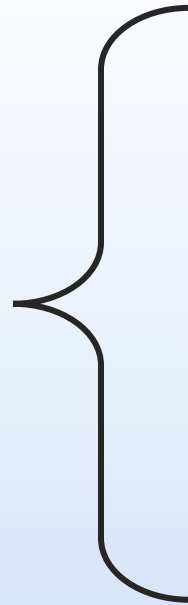
- Encounter trends for “Some Other Race” and Black/African American patients closely align over time
- Multiple Non-White groups respond to system expansion in parallel at a similarly lower growth scale

Stage-Setting Context: Income Inequality in Kansas



- **Higher median income in Kansas.**
- **Stable employment employers sponsor health insurance.**
- **Inflation.**

Thank you



We sincerely appreciate your
time and attention to this
presentation.